Zero Shot Sentiment Analysis on Tweets in Any Language

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Objectives

We will build a sentiment analysis [1] convolutional neural network classification model, pretrained on the geolocation problem, by training it on a set of sentiment-labeled tweets in a popular language. We aim to zero shot learn to classify sentiment in any other language by training on a set of unlabeled tweets in that language.

Introduction

We aim to leverage the abundance of geolocation data, as well as the Unicode-level similarity, of tweets across different languages to learn to classify sentiment in a target *low resource language* given labeled data in a *high resource language* and unlabeled data in the target language.

(a) English (b) Portuguese

Figure: Similar in location and Unicode representation

We will compare the results against a model trained on tweets in a target high resource language from the [2] dataset, whose precision is shown below.

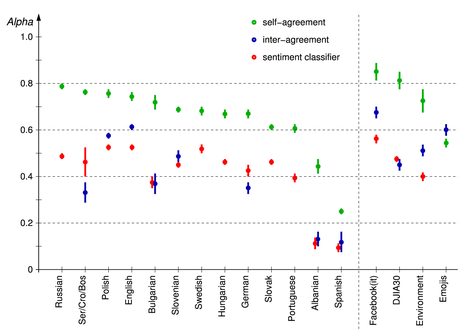


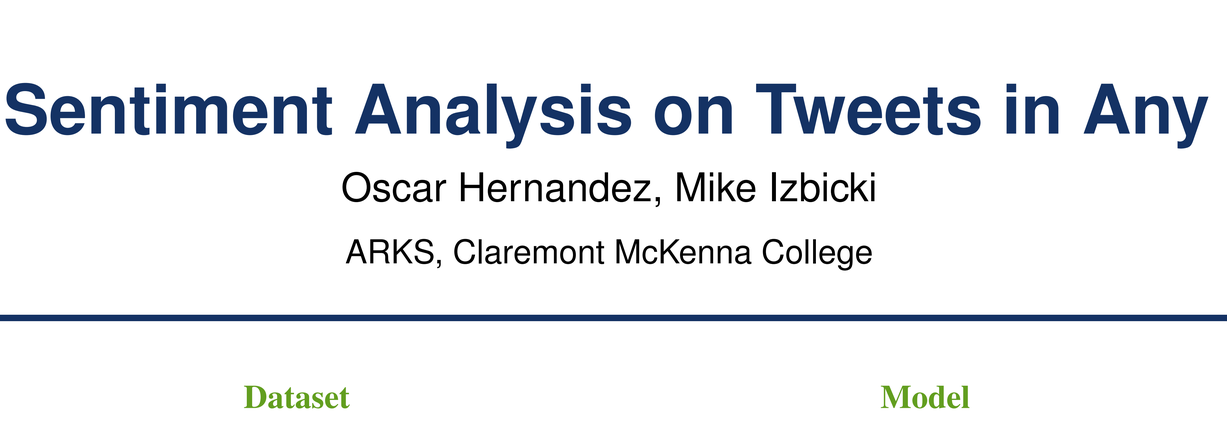
Figure: Label agreement across annotators

Dataset

We use a publicly available sentiment-labeled dataset [2] containing 1.6 million tweets in 13 languages. Since all existing sentiment classifiers require labeled data, this allows us to train models on many languages besides English, which accounts for 42% of tweets. However, there are still hundreds of languages popular on Twitter with insufficient amounts of sentiment-labeled data.
However, repeating this process to obtain more data for all other 100 languages actively used on Twitter [3] is impractical, and likely yield data equally or worse in quality.

Model

In this experiment, we will use a UnicodeCNN pretrained to solve the geolocation problem all 900 million geo-encoded tweets sent over 9 months by over 3.0 million users in over 100 languages, and replace the final MvMF layer with a layer of sentiment classification as shown below. We will retrain some layers of the model on a training set and evaluate on a testing set, where the languages of the two sets may or may not agree.
We choose to run our experiments on English and Portuguese, which have a large amount of high-quality sentiment-labeled data.



UnicodeCNN

Inspired by the *Character-Level Convolutional Neural Network* (CLCNN) which classifies English text by using 70 characters commonly found in English text, the *Unicode Convolutional Neural Network* (UnicodeCNN) [4] is a state of the art model for tweet geolocation which predicts the exact GPS coordinates of any tweet in any language at any location [5].

Experiments

#	Retrain	Train	Test
1	All	English	English
2	All	Portuguese	Portuguese
3	All	English	Portuguese
4	Final	English	English
5	Final	Portuguese	Portuguese
6	Final	English	Portuguese

Figure: Main experiments highlighted.

Contribution

In the main experiments, we train on sentiment-labeled English tweets, test on unlabeled Portuguese tweets, and validate against a model trained on sentiment-labeled Portuguese tweets. Since our tests did not require the labels, we would expect its success to imply the the English-trained model would also succeed on low resource languages. After finalizing the results on Portuguese, we would like to test and validate this model on tweets in other languages.

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